INVESTIGATING SPATIAL STRUCTURES in Urban Environments FOR LEGAL AND POLICY INSIGHTS LAB

Enhancing Brain Injury Surveillance Using Large Language Models and Topic Modeling

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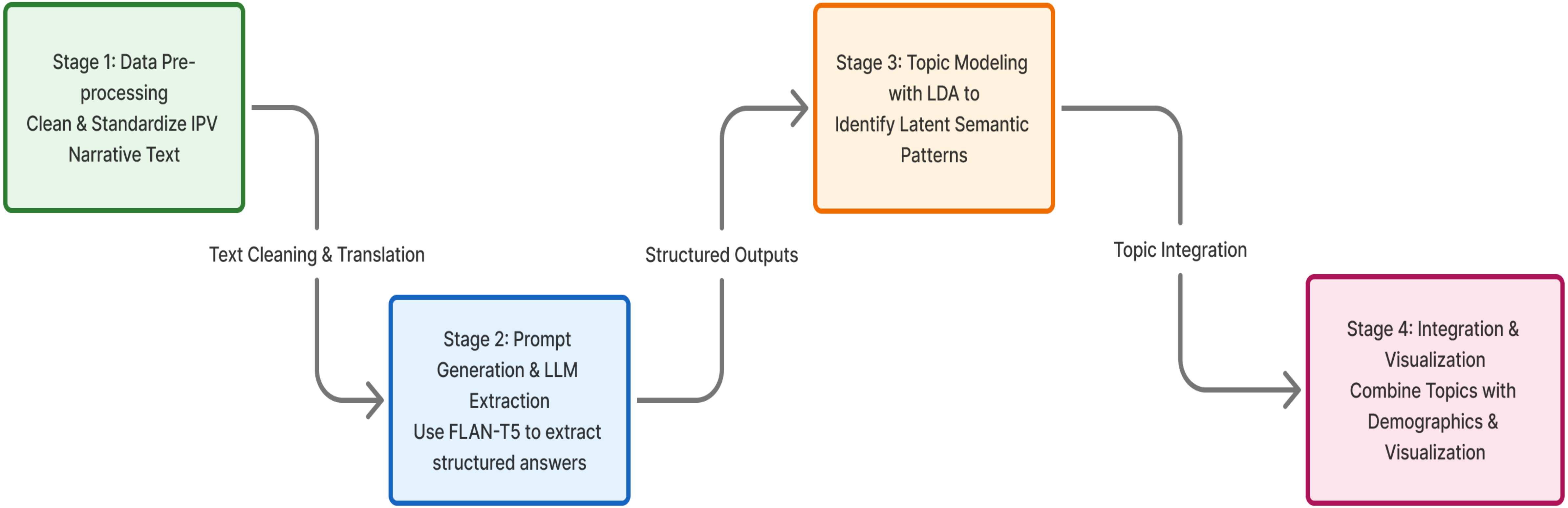
Background

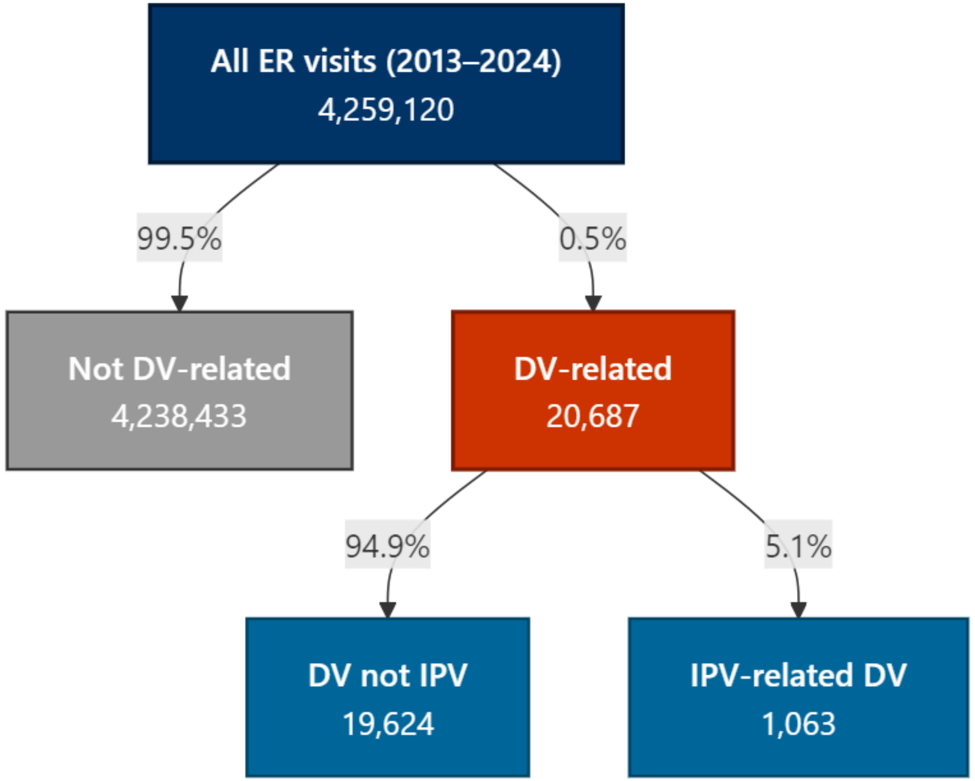
- Head and neck trauma are among the most common and severe injuries among survivors of intimate partner violence (IPV), resulting in traumatic brain injury (TBI)
- Existing surveillance systems rarely distinguish the underlying mechanisms of TBI in IPV, limiting insight into causal mechanisms and prevention
- The National Electronic Injury Surveillance System (NEISS) captures both structured injury codes (e.g., body part, diagnosis) and unstructured free-text narratives that contain rich contextual details about injury circumstances
- The NEISS narrative fields have been used for <<see our paper>>, but they remain underutilized for identifying violence-related mechanisms or typologies of harm

AI-Assisted Brain Injury Surveillance: Study Goals & Analytical Framework

- **Goal:** To develop an artificial intelligence (AI) pipeline that applies a large language model (LLM) to extract and classify behavioral and contextual information from NEISS IPV narratives.
- **Aim:** To identify and categorize head and neck injuries associated with IPV-related harm by integrating narrative-derived features with structured NEISS variables (e.g., diagnosis and injury location).



Identifying Intimate Partner Violence (IPV) in NEISS Narratives



- NEISS narratives contain both explicit (e.g., “domestic violence? And implicit indicators of conflict, aggression, or self-harm that are captured by the structured variables
- We created regular expression routines to flag narratives containing violence
  - We first used vectorized regex functions to remove common descriptors and expand abbreviations (e.g., lac = laceration) to transform the narratives to a human-readable format
  - Within violence narratives, cases referencing relationship terms (e.g., *boyfriend, girlfriend, husband, wife, partner*) were identified as Intimate Partner Violence (IPV)
  - IPV-related terms but excluding incidents involving intimates that were not IPV (e.g., accidental self-injury, agitation, and play fighting)

```
dict_abb_exp <- c(
  'pt'='patient','c/o'='complains of','lac'='laceration','rt'='right','l'='left','r'='right',
  's/p'='status post','w/'='with','@'='at','fx'='fracture','loc'='loss of consciousness',
  'ed'='emergency department','sts'='states','acc'='accidentally','w'='with','hx'='history',
  'hd'='head','bib'='brought in by','lt'='left','inj'='injury','er'='emergency room',
  'bal'='blood alcohol level','fb'='foreign body','w/o'='without','pn'='pain',
  'nh'='nursing home','co'='complains of','p/w'='presents with','lwr'='lower',
  'str'='strained','spr'='sprain','sust'='sustained','h/o'='history of',
  "s'd&f"='slipped, tripped, and fell',"t'd&f"='tripped and fell','bwd'='backward'
)
```

We generated custom prompts from the unstructured narratives by merging them with the structured variables to create human-readable text descriptions of the narratives (e.g., *patient, after argument w/ boyfriend, was locked out of house. tried to get into home by breaking glass window. +ethyl alcohol. clinical diagnosis: arm lacerations*).
Each prompt was then followed by a small set of targeted questions designed to capture key contextual and clinical details of each event.

```
# Cleaned narrative
narrative = str(row.get(TEXT_COL, "")) if pd.notna(row.get(TEXT_COL, "")) else ""

# Structured variables for context
age = row.get("age", "")
sex = str(row.get("sex", "")).lower()
diagnosis = str(row.get("diagnosis", "")).lower()
body_part = str(row.get("body_part", "")).lower()
disposition = str(row.get("disposition", "")).lower() if "disposition" in row else ""
location = str(row.get("location", "")).lower() if "location" in row else ""

# Expanded prompt
structured_context = (
    f"This record describes a {age}-year-old {sex} patient presenting to the emergency department. "
    f"The reported diagnosis was {diagnosis} affecting the {body_part}. "
    f"The patient was {disposition}. "
    f"The incident occurred at {location}. "
    f"Below is the clinician or patient description of the incident:\n\n"
    f"{narrative.strip()}"
)
```

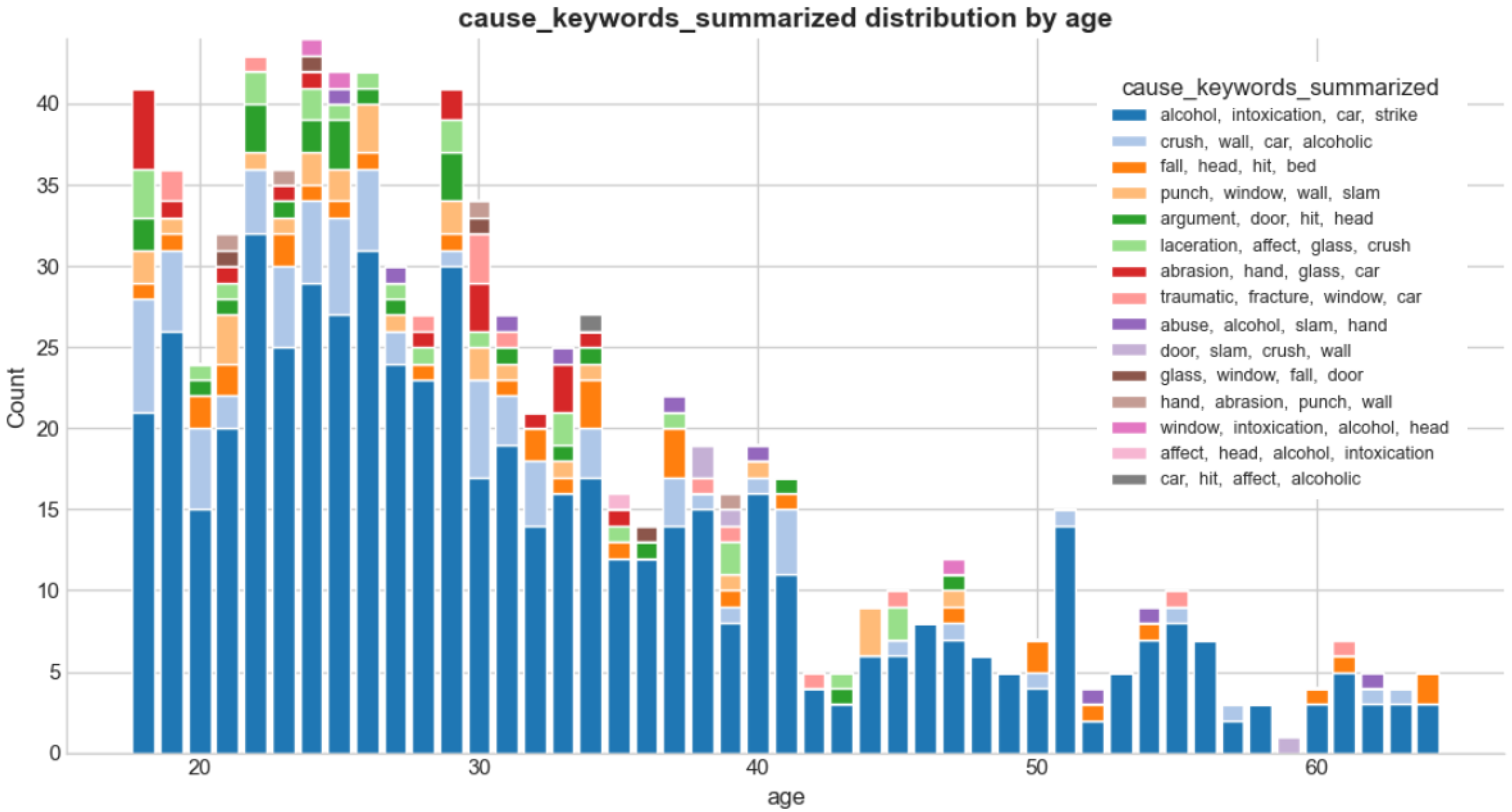
LLM → LDA Output	**Text**
Raw NEISS Narrative	21YOF REPORTS SHE GOT INTO AN ARGUMENT W/ HER BOYFRIEND AND GRABBED A KITCHEN KNIFE AND PUT IT TO HER THROAT STATING SHE WAS GOING TO SLIT HER THROAT. PT STATES THIS WAS A MISTAKE AND SHE ACTED IMPULSIVELY IN THE MOMENT. CURRENTLY HAS NO SUICIDAL IDEATIONS OR SUICIDAL PLAN. CANNABINOID+, BAL <10. DX: SUICIDAL BEHAVIOR W/O ATTEMPTED SELF INJURY.
Cleaned Narrative Sent to LLM (Contextual prompt)	patient reports she got into an argument with her boyfriend and grabbed a kitchen knife, putting it to her throat while stating she was going to slit her throat. patient states this was a mistake and she acted impulsively in the moment. currently has no suicidal ideation or plan. cannabinoid positive, blood alcohol level less than 10. clinical diagnosis: suicidal behavior without attempted self-injury.\n\nQuestion: What was the person doing before the injury occurred?
LLM-Generated Structured Output	**action:** “arguing with boyfriend” **cause:** “grabbed kitchen knife and held it to throat” **diagnosis:** “suicidal behavior without self-injury” **how:** “impulsive self-harm gesture following argument” **what:** “kitchen knife” **where:** “home”

From LLM to LDA

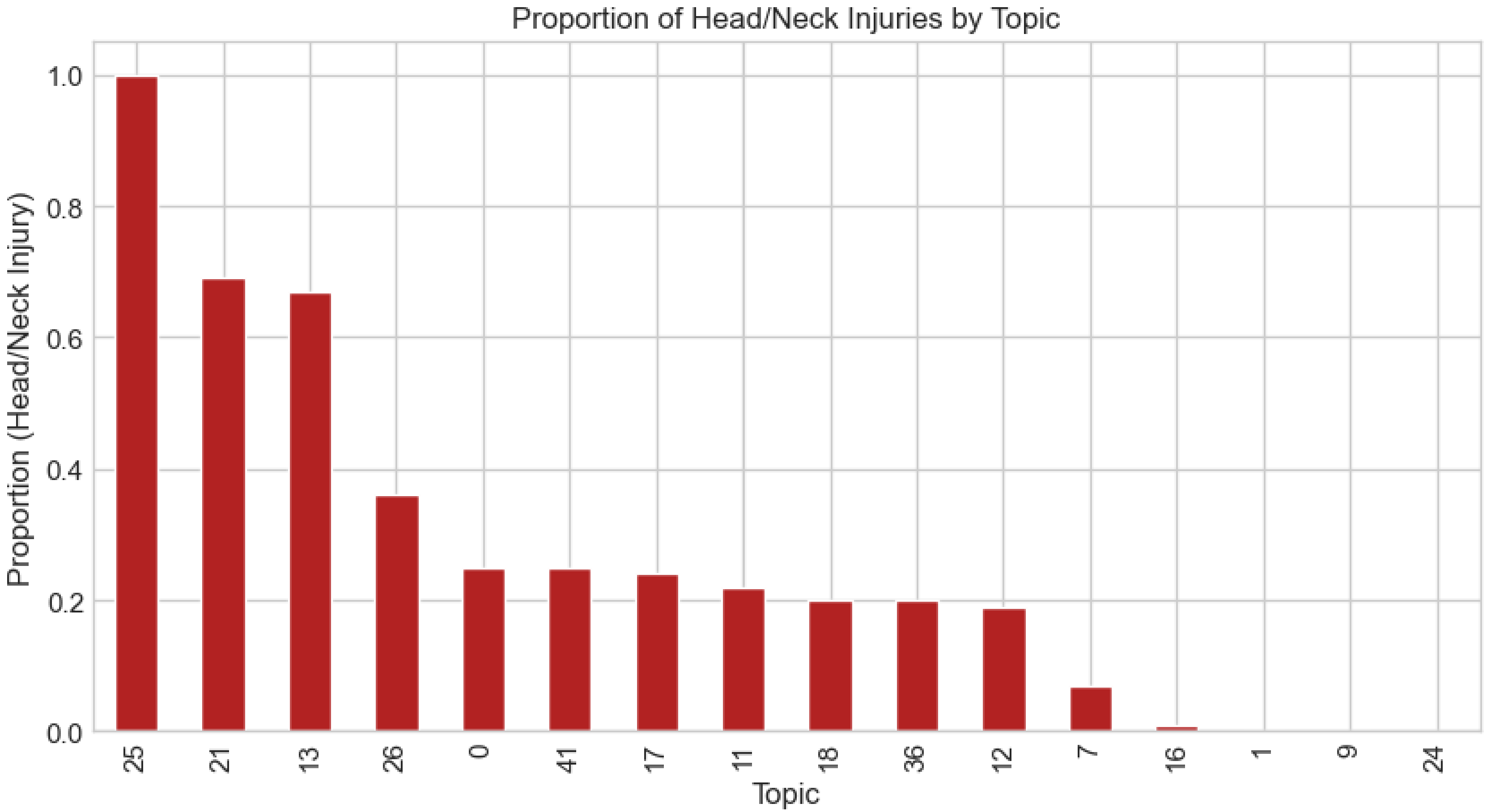
- We converted the unstructured free text into six targeted semantic representations of the same event.
- Each LDA model runs within one question column, discovering themes across all responses to that specific question
  - **Latent Cause** column → entries like *"wall and TV," "electric shock," "fall from stairs“*
  - **Interaction/Action** column → entries like *"arguing with boyfriend," "slipped and fell“*
  - **Medical outcomes/Diagnosis** column → entries like *"contusion," "fracture," "closed head injury"*
- Each LDA topic converted into a short label and visualized across demographic groups to assess differences in topic prevalence

action	cause	diagnosis	how	what	where
[arguing, with, boyfriend]	[hematoma]	[closed, head, injury]	[the, patient, was, arguing, with, boyfriend]	[slipped]	[carpet]
[in, an, argument, with, his, girlfriend]	[wall, and, tv]	[left, hand, contusion]	[the, patient, hit, wall, and, tv, with, fists]	[he, fell]	[wall]
[punching, window]	[electric, shock]	[hand, laceration]	[electric, shock]	[electric, shock]	[window]
[she, was, slammed, down, on, the, counter]	[shot, glass]	[the, patient, has, laceration, to, its, finger, cut]	[the, patient, slammed, shot, glass, down, on, the, counter, at, home, angry, with, husband]	[she, fell]	[on, the, counter]
[an, argument, with, boyfriend]	[an, argument, with, boyfriend]	[ear, ache]	[an, argument, with, boyfriend]	[he, shook, gas, can, at, her]	[on, its, foot]

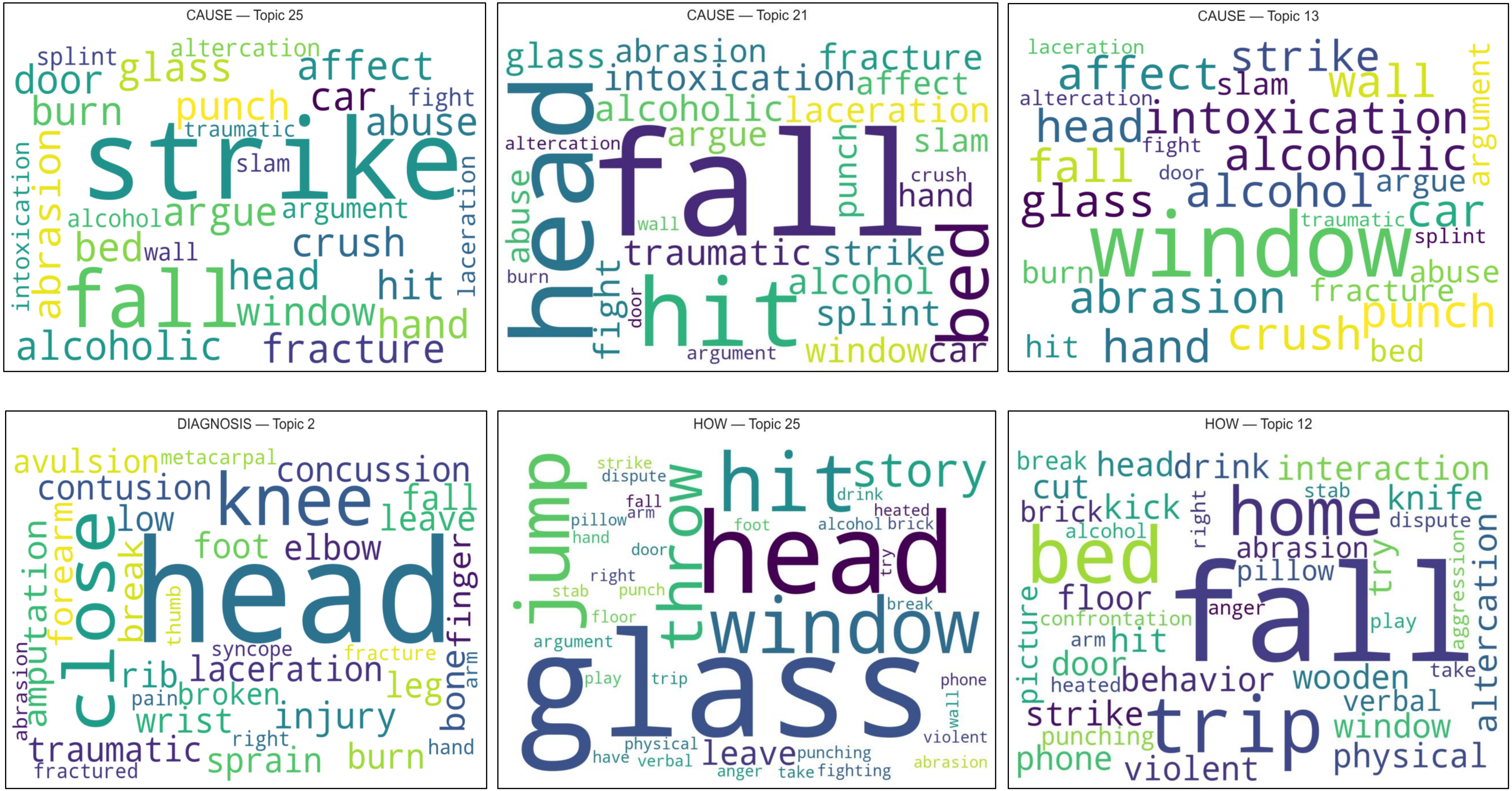
Dimension	Top Words	Summary Interpretation
Action	argue, play, punch, verbal, wall, fight	Reflects interpersonal conflict and self-directed aggression during verbal altercations, often escalating to physical contact.
Cause	alcohol, intoxication, car, strike, window, glass	Indicates alcohol-related and blunt impact injuries, including falls and collisions linked to intoxication.
Diagnosis	hand, fractured, broken, injury, head, finger	Shows predominance of musculoskeletal and soft-tissue trauma, particularly fractures and contusions to the upper body.
How	behavior, play, fighting, punch, violent, interaction	Captures injuries arising from interpersonal struggle or impulsive acts during emotionally charged events.
What	wall, glass, window, brick, door, punch	Highlights impacts with household structures or objects commonly involved in physical altercations.
Where	home, outside, public, place, bathroom	Suggests most incidents occurred in the home or outside the home, with some in public or outdoor settings.


cause_keywords_summarized distribution by age
Count
age
0
5
10
15
20
25
30
35
40
20
30
40
50
60
cause_keywords_summarized
alcohol, intoxication, car, strike
crush, wall, car, alcoholic
fall, head, hit, bed
punch, window, wall, slam
argument, door, hit, head
laceration, affect, glass, crush
abrasion, hand, glass, car
traumatic, fracture, window, car
abuse, alcohol, slam, hand
door, slam, crush, wall
glass, window, fall, door
hand, abrasion, punch, wall
window, intoxication, alcohol, head
affect, head, alcohol, intoxication
car, hit, affect, alcoholic

From LLM to LDA



- Next, we identified which injury records are head/neck injuries and filtered the data to keep only those cases
- We got the prevalence of head/neck injury within each topic


CAUSE — Topic 25
splint
altercation
door
glass
affect
burn
punch
car
fight
traumatic
abuse
strike
intoxication
abrasion
slam
laceration
alcohol
argue
argument
bed
wall
crush
fall
head
hit
window
hand
alcoholic
fracture
CAUSE — Topic 21
glass
abrasion
fracture
intoxication
affect
head
alcoholic
laceration
altercation
argue
slam
fall
punch
crush
abuse
hand
wall
traumatic
strike
bed
burn
fight
alcohol
door
hit
splint
argument
window
car
CAUSE — Topic 13
laceration
strike
affect
slam
wall
argument
altercation
head
intoxication
fight
alcoholic
fall
door
alcohol
argue
glass
car
traumatic
window
splint
burn
abuse
fracture
abrasion
punch
crush
hit
hand
bed
DIAGNOSIS — Topic 2
avulsion
metacarpal
concussion
contusion
knee
fall
low
leave
forearm
foot
elbow
close
amputation
break
head
finger
thumb
syncope
fracture
abrasion
rib
laceration
leg
arm
pain
broken
bone
wrist
injury
right
traumatic
hand
sprain
burn
fractured
HOW — Topic 25
strike
dispute
hit
story
jump
throw
drink
fall
arm
heated
pillow
foot
alcohol
brick
hand
head
door
try
right
stab
punch
floor
break
window
argument
glass
play
trip
phone
wall
violent
physical
have
verbal
leave
punching
abrasion
anger
take
fighting
HOW — Topic 12
break
head
drink
interaction
cut
stab
knife
brick
kick
right
home
alcohol
dispute
bed
abrasion
try
pillow
aggression
altercation
floor
anger
fall
confrontation
play
picture
arm
hit
door
take
heated
behavior
wooden
strike
trip
verbal
window
punching
phone
violent
physical

Injury Typology	%	Diagnosis Codes	Body Part Codes	Cause Topics	Representative Narrative
Accidental fall or slip	38.6%	52 (Concussion), 59 (Laceration), 62 (Closed-head injury)	75 (Head), 79 (Hip/Back), 76 (Face)	18, 21, 26	*“Patient was arguing with boyfriend, slipped and fell backward hitting head on carpet. Diagnosis: closed-head injury.”*
Self-inflicted (intentional self-harm)	19.9%	53 (Contusion), 59 (Laceration), 62 (Head injury), 57 (Fracture)	75 (Head), 79 (Back), 76 (Face)	18, 21, 26, 13, 17	*“Patient began banging head against wall during argument with husband. Diagnosis: closed-head injury.”*
Physical assault (partner-inflicted)	25.1%	59 (Laceration), 62 (Closed-head), 57 (Fracture), 71 (Pain)	75 (Head), 76 (Face), 79 (Abdomen)	18, 21, 12, 17	*“Patient was in argument with wife and she stabbed him in the neck with kitchen knife. Diagnosis: stab wound, neck.”*
Blunt object impact	9.9%	59 (Laceration), 62 (Closed-head), 53 (Contusion), 52 (Concussion)	75 (Head), 76 (Face), 85 (Multiple sites)	18, 17, 21	*“Patient got angry at girlfriend and head-butted wall. Diagnosis: forehead abrasion, forehead hematoma.”*
Environmental or chemical	2.3%	71 (Pain/irritation), 74 (Dermatitis)	76 (Face), 88 (Mouth), 94 (Ear)	18	*“Patient sprayed self with pepper spray while trying to spray boyfriend during fight. Diagnosis: contact dermatitis.”*
Vehicle-related impact	1.7%	62 (Head injury), 57 (Fracture)	75 (Head), 79 (Back), 83 (Foot)	18, 36	*“Patient in argument with fiancé swerved and crashed into pole, complains of head and neck pain.”*

Significance and Future Directions

- **Key Insights**
  - Combining regex typology (mechanism-based classification) and topic modeling (semantic clustering) enables a detailed mapping of *how, where, and why* brain injuries occur during an IPV-related incident.
  - The AI-assisted pipeline transforms unstructured clinical narratives into quantitative data describing mechanisms of harm.
- **Public Health Relevance**
  - Many “accidental” or “environmental” injuries co-occur with relationship conflict, revealing blurred boundaries between unintentional and interpersonal harm.
  - Enhances injury surveillance and improves early detection of violence.
  - Supports data-driven prevention and policy design targeting interpersonal and domestic injury risk.
- **Next Steps**
  - Validate typologies with larger datasets.
  - Expand LLM prompts to capture more contextual cues.
  - Examine patterns across sociodemographic and clinical subgroups.